Connor **Reed**

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Education

New York University *New York, NY*
Master of Science, Data Science *2022*

Relevant Coursework: Probability & Statistics for Data Science • Optimization & Computational Linear Algebra • Machine Learning • Deep Learning • Inference & Representation • Forecasting Time Series Data

Yale University *New Haven, CT*
Bachelor of Science, Environmental Studies *with distinction, cum laude* *2019*

- *Thesis title*: Engaging open-source precision viticulture to manage spatial heterogeneity and improve cover-cropping practice on an organic vineyard (Abstract)

School for Field Studies //
Ugyen Wangchuck Institute for Conservation and Environmental Research *Bumthang District, Bhutan*
Study Abroad *June – July 2017*

- Completed coursework on Himalayan forest ecology, rural livelihoods, and Bhutanese culture
- Conducted ethnographic field research on agricultural knowledge transfer in the village of Ura and presented findings to government stakeholders

Research Experience

New York University, Department of Environmental Studies *New York, NY*
Graduate Research Assistant, McDermid Lab *July 2020 – September 2022*

- Developed end-to-end geospatial deep learning pipeline to infer flood extent and smallholder cropland distribution from Sentinel-1 SAR and Sentinel-2 multispectral time series across sub-Saharan Africa
- Published geospatial causal analysis localizing and quantifying the impact of flood events on food security outcomes (IPC rating) across sub-Saharan Africa from 2009–2020 using panel regression and Granger causality testing [1]

Yale School of the Environment *New Haven, CT*
Research Assistant, The Bradford Lab *January 2016 – August 2020*

- Conducted meta-analysis (data collection, statistical analysis) of long-term experiments to quantify the mediating effect of soil organic carbon on the relationship between conservation agriculture practices and crop yield stability

Publications

[1] **C. Reed** *et al.*, "The impact of flooding on food security across Africa," *Proceedings of the National Academy of Sciences*, vol. 119, no. 43, p. e2119399119, 2022, doi: 10.1073/pnas.2119399119.

Professional Experience

Aquabyte *New York, NY*
Senior Machine Learning Engineer *December 2024 – present*
Machine Learning Engineer *June 2022 – December 2024*

- Researched and deployed a multi-model sea lice detection pipeline for ecological monitoring, operating on underwater camera imagery of farmed fish: YOLOv9 detects candidate lice on image tiles; a custom transformer—designed from neural field principles—jointly encodes the full image and detected bounding boxes into a shared latent space to classify each detection by anatomical fish region; a companion multilabel quality model built on DINOv3-ViT gates predictions by per-region image fidelity
- Designed a statistical pipeline combining model detections with small sample of human QA to generate daily, unbiased population-level estimates of sea lice burden per fish; validated against field measurements, establishing a scalable biomonitoring framework deployed across 400+ pens
- Developed a 16-day recursive forecasting model of pen-level sea lice population dynamics using LightGBM with graph-engineered spatial features and counterfactual intervention simulation enabling optimization of treatment timing across farm pen networks

Indigo *Boston, MA*
Soil Data Research Intern, Carbon Experimentation *June – August 2021*

- Developed generative Bayesian models to synthesize soil datasets from heterogeneous published and proprietary sources, enabling simulation of soil carbon responses to management practices across diverse geographies
- Performed quantitative analysis of carbon sequestration potential and risk-reward tradeoffs across agricultural land use scenarios to identify high-value carbon market opportunities

Grants & Honors

2019 Gaylord Donnelley Prize in Environmental Studies, Yale University, Department of Environmental Studies

2019 Yale Global Food Fellowship, Yale Sustainable Food Program

2018 Yale Summer Environmental Fellowship, Yale University, Department of Environmental Studies

2017 Yale Global Food Fellowship, Yale Sustainable Food Program

2017 **Robert Berlin Fellowship**, Yale University, Berkeley College

2017 **Tristan Perlroth Prize for Summer Foreign Travel**, Yale University

2016 **Yale College First-Year Summer Research Fellowship in the Sciences & Engineering**, Yale University

Skills

Programming Python, SQL, Bash
ML Tools PyTorch, scikit-learn, NumPy, pandas, ONNX, TensorRT, FiftyOne, Hydra, MLFlow, Weights & Biases
Geospatial Google Earth Engine, GeoPandas, rasterio, xarray, GDAL
Databases PostgreSQL, Snowflake
DevOps Git, Docker, Terraform, dbt, Airflow, AWS
Statistics Machine learning, deep learning, computer vision, time series, probabilistic models, Bayesian inference, hypothesis testing, A/B testing, generative models
Other Research, writing, public speaking